

Notes on MATH 223

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Chapter 1. Complex Numbers

1. Structure of Complex Numbers

Definition 1.1.1 (Complex Numbers):

The set of **complex numbers**, denoted by $\mathbb{C}$, is the Cartesian product of the real numbers with itself. That is

$$\mathbb{C} := \mathbb{R} \times \mathbb{R}.$$

A typical element $z \in \mathbb{C}$ can be expressed as an ordered pair of real numbers (x, y) . The set of complex numbers as for now is just a set of ordered pairs. Thus, we shall define some operations on $\mathbb{C}$ to give it more structure.

Definition 1.1.2 (Operations Within Complex Numbers):

The **addition** of two complex numbers $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$ is defined as

$$z_1 + z_2 := (x_1 + x_2, y_1 + y_2).$$

The **multiplication** of two complex numbers $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$ is defined as

$$z_1 z_2 := (x_1 x_2 - y_1 y_2, x_1 y_2 + x_2 y_1).$$

Theorem 1.1.3 (Complex Field):

The set of complex numbers $\mathbb{C}$ with the operations of addition and multiplication defined in **Definition 1.1.2** are *well-defined* and *satisfy the properties that we expect them to have*.

Definition 1.1.4 (Imaginary Unit):

The **imaginary unit** i is defined as the complex number $(0, 1)$.

The imaginary unit i has a very interesting property, which is that

$$i^2 = (0, 1)(0, 1) = (-1, 0).$$

Corollary 1.1.5 (Decomposition of Complex):

A complex number $z = (x, y)$ can be written as $z = (x, 0) + i(y, 0)$.

Proof 1.1.5:

Consider a complex number $z = (x, 0) + i(y, 0)$. We have

$$z = (x, 0) + (0, 1)(y, 0) = (x, 0) + (0, y) = (x, y).$$

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Theorem 1.1.6 (Embedding of Real in Complex):

Let R be the set of all complex numbers of the form $(x, 0)$, where $x \in \mathbb{R}$. That is

$$R = \{(x, 0) : x \in \mathbb{R}\}.$$

Then, R is a subset of $\mathbb{C}$ and R is *structurally equivalent* to $\mathbb{R}$.

Proof 1.1.6:

As we need to properly define the word *equivalent* before we can prove the theorem, we shall only first do an informal demonstration of the theorem here.

To show that R is *structurally equivalent* to $\mathbb{R}$, let's first show that R is *itself equivalent* to $\mathbb{R}$. Consider an element $(x, 0) \in R$, we can associate it to a **unique** element $x \in \mathbb{R}$ and **vice versa**. This is called a **bijection**, which somewhat means that R and $\mathbb{R}$ have the same number of elements.

Furthermore, we claim that the **addition** and **multiplication** of two elements in R are *equivalent* to the addition and multiplication of the corresponding elements in $\mathbb{R}$. To show this, consider two elements $(x_1, 0), (x_2, 0) \in R$, we have

$$(x_1, 0) + (x_2, 0) = (x_1 + x_2, 0),$$

which corresponds to the sum of x_1 and x_2 in $\mathbb{R}$. Similarly, we have

$$(x_1, 0)(x_2, 0) = (x_1 x_2, 0),$$

which also corresponds to the product of x_1 and x_2 in $\mathbb{R}$.

Since the **addition** and **multiplication** on these two sets are *equivalent*, any other operations derived from these two operations (such as subtraction, division, exponential) are also *equivalent*. Hence, we can conclude that R and $\mathbb{R}$ are *structurally equivalent*.

Convention 1.1.7:

Now, due to this *structural equivalence*, we will denote any complex number of the form $(x, 0)$ directly by x and **call** this number a **real number**.

Moreover, we will borrow any function $f : \mathbb{R} \mapsto \mathbb{R}$ to R , by defining a function $f_R : R \mapsto R$ as

$$f_R((x, 0)) := (f(x), 0).$$

Corollary 1.1.8 (Alternative Notation):

A complex number $z = (x, y)$ can be written as

$$z = (x, y) = (x, 0) + i(y, 0) = x + iy.$$

For a complex number $z = x + iy$, we call x the **real part** of z and denote it by $\Re(z)$ or $\text{Re}(z)$, and we call y the **imaginary part** of z and denote it by $\Im(z)$ or $\text{Im}(z)$.

Definition 1.1.9 (Complex Conjugate):

The **complex conjugate** of a complex number $z = x + iy$ is defined as

$$\bar{z} := x - iy.$$

One can immediately verify that the complex conjugate has the following property:

$$z\bar{z} = (x + iy)(x - iy) = x^2 + y^2.$$

Due to **Convention 1.1.7**, we treat $x^2 + y^2$ as real number. Since $z\bar{z} = x^2 + y^2$ is a **non-negative real number**, we can take the square root of it.

Definition 1.1.10 (Modulus):

The **modulus** of a complex number $z = x + iy$ is defined as

$$|z| := \sqrt{z\bar{z}} = \sqrt{x^2 + y^2}.$$

Corollary 1.1.11 (Modulus and Conjugate):

The modulus of a complex number z is always a non-negative real number, that is,

$$|z| \geq 0.$$

Corollary 1.1.12 (Inverse of a Complex Number):

A complex number $z = x + iy \neq 0$ has a multiplicative inverse, which is given by

$$\frac{1}{z} = \frac{1}{z} \frac{\bar{z}}{\bar{z}} = \frac{\bar{z}}{|z|^2} = \frac{x}{|z|} - i \frac{y}{|z|}$$

2. Polynomials and Complex Roots

Theorem 1.2.1 (Fundamental Theorem of Algebra):

Every non-constant polynomial with complex coefficients has at least one complex root.

Theorem 1.2.2 (Factorization of Polynomials):

If f is a degree n polynomial with complex coefficients, then f can be factored into n **linear factors**.

$$f = a(X - z_1)(X - z_2)\dots(X - z_n),$$

where $z_1, z_2, \dots, z_n$ are the complex roots of f and a is the leading coefficient of f .

3. Geometric Interpretation of Complex Numbers

Lemma 1.3.1 (Euler's Formula):

Let $\theta \in \mathbb{R}$. Then, we have

$$e^{i\theta} = \cos(\theta) + i \sin(\theta).$$

Corollary 1.3.2 (Modulus of Exponential):

One can immediately verify that the modulus of $e^{i\theta}$ is always 1, that is,

$$|e^{i\theta}| = \sqrt{\cos^2(\theta) + \sin^2(\theta)} = 1.$$

Corollary 1.3.3 (Polar Form):

One can also verify that any complex number z can be written in the form

$$z = re^{i\theta},$$

where $r = |z| \in \mathbb{R}$ and $\arg(z) := \theta$ is called the **argument** of z .

Chapter 2. Algebraic Structures

1. Groups, Rings, and Fields

Definition 2.1.1 (Groups):

A **group** is a set G together with a binary operation $\cdot$ that satisfies the following properties:

1. **Closure:** For all $a, b \in G$, we have $a \cdot b \in G$.
2. **Associativity:** For all $a, b, c \in G$, we have $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
3. **Identity:** There exists an element $e \in G$ such that for all $a \in G$, we have $e \cdot a = a \cdot e = a$.
4. **Inverse:** For each element $a \in G$, there exists an element $b \in G$ such that $a \cdot b = b \cdot a = e$.

Definition 2.1.2 (Monoids and Semigroups):

A **monoid** is a set S together with a binary operation $\cdot$ that satisfies the following properties:

1. **Closure**
2. **Associativity**
3. **Identity**

A **semigroup** is a set S together with a binary operation $\cdot$ that satisfies the following properties:

1. **Closure**
2. **Associativity**

Corollary 2.1.3 (Uniqueness of Identity and Inverse):

The **identity element** and the **inverse of each element** in a group are unique.

Proof 2.1.3:

Let G be a group and let e and e' be two identity elements in G . Then, we have

$$e = e \cdot e' = e'.$$

Hence, the identity element is unique.

Let $a \in G$ and let b and b' be two inverses of a . Then, we have

$$b = b \cdot e = b \cdot (a \cdot b') = (b \cdot a) \cdot b' = e \cdot b' = b'.$$

Hence, the inverse of each element is unique.

Definition 2.1.4 (Abelian Groups):

A group G is called an **Abelian group** if the binary operation $\cdot$ is **commutative**, that is, for all $a, b \in G$, we have $a \cdot b = b \cdot a$.

Definition 2.1.5 (Rings):

A **ring** is a set R together with two binary operations $+$, called **addition**, and $\cdot$, called **multiplication**, such that:

1. $(R, +)$ is an **Abelian group**.
2. $(R, \cdot)$ is a **monoid**.
3. **Distributive Laws:** For all $a, b, c \in R$, we have
 - $a \cdot (b + c) = a \cdot b + a \cdot c$
 - $(a + b) \cdot c = a \cdot c + b \cdot c$

Definition 2.1.6 (Commutative Rings):

A **commutative ring** is a ring $R(+, \cdot)$ such that the **multiplication is commutative**, that is, for all $a, b \in R$, we have $a \cdot b = b \cdot a$.

Definition 2.1.7 (Fields):

A **field** is a set F together with two binary operations $+$ and $\cdot$, with its **additive identity** denoted by 0 , such that:

1. $F(+, \cdot)$ is a **commutative ring**.
2. Every element $f \in F$ such that $f \neq 0$ has a **multiplicative inverse** in F .

Theorem 2.1.8 (Real Field):

The set of real numbers $\mathbb{R}$ with the usual addition and multiplication forms a **field**.

Proof 2.1.8:

Let's start by showing that $\mathbb{R}(+)$ is an Abelian group.

1. **Closure:** For all $a, b \in \mathbb{R}$, we have $a + b \in \mathbb{R}$.
2. **Associativity:** For all $a, b, c \in \mathbb{R}$, we have $(a + b) + c = a + (b + c)$.
3. **Identity:** The additive identity is 0 , since for all $a \in \mathbb{R}$, we have $0 + a = a + 0 = a$.
4. **Inverse:** For each element $a \in \mathbb{R}$, the additive inverse is $-a$, since we have $a + (-a) = (-a) + a = 0$.
5. **Commutativity:** For all $a, b \in \mathbb{R}$, we have $a + b = b + a$.

Now, let's show that $\mathbb{R}(\cdot)$ is a monoid.

1. **Closure:** For all $a, b \in \mathbb{R}$, we have $a \cdot b \in \mathbb{R}$.
2. **Associativity:** For all $a, b, c \in \mathbb{R}$, we have $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
3. **Identity:** The multiplicative identity is 1 , since for all $a \in \mathbb{R}$, we have $1 \cdot a = a \cdot 1 = a$.

Due to the **distributive laws** of real numbers, we can conclude that $\mathbb{R}$ is a **ring**. Moreover, it is a **commutative ring** since the multiplication of real numbers is commutative. Finally, **every non-zero real number has a multiplicative inverse**, which is also a real number. Hence, we can conclude that $\mathbb{R}$ is a **field**.

Theorem 2.1.9 (Complex Field):

Now, we can formally restate **Theorem 1.1.3** as follows: The set of complex numbers $\mathbb{C}$ with the operations of addition and multiplication defined in **Definition 1.1.2** forms a **field**.

The proof of this theorem follows the same idea as the proof for **Theorem 2.1.8**.